

Paul that was a good set of questions. Oh my gosh. It's amazing. We you know, we learned a lot about sponges and webs and all sorts of things

Coming up another installment of Cosmic Query's grab bag additional

Welcome to Star Talk your place in the universe where science and pop culture collide Star Talk begins right now This is Star Talk Cosmic Query's edition this time I got Paul MacCurio Paul welcome back dude. Hi. Good to see you again Neil How are you buddy? All right? I Paul at some point somebody christened you Baron Paul MacCurio

That is true. And you had a we had a ceremony which was okay in your office if you remember you did we use my Excalibur store. Yes Yeah, I still for some reason there's no red carpet here, but you know, you you still have your traveling show permission to speak I do. Yeah, there's a theater show, right? It's a it's a wagon and I sell medicines I'm a medicine man now because I'm gonna be out of a job at the late show So yes, touring with that. I've got my podcast inside out with Paul MacCurio because it's the law and everyone's supposed to have a

podcast now and You've been in the biz a while Yeah, I go back to I go back to the daily show and we one of the original guys on that and But even pre-dating John Stewart John Stewart that tiny little man. Yes

He's only this big a lot of people don't know the cameras are amazing that that we would have yes with Craig Kilbourne and then and Lewis Black was on and we would have all this crazy fun and Nobody knew what they were doing, but they were letting us we were like, you know You would appreciate this we were like in the in the lab Just they just said just go make it and blow things up

And then we get yelled at by the president of the network every once in a while because we went too far He would literally yell at you like your father. We yell at you like that. He couldn't get

We got a grab bag this session. Yeah, so that means great questions by the way some really good questions Whatever subjects just sort of landed in

your lap and these are questions asked by our patreon members and whatever I do these I'm always and this isn't false praise like you're the fans of our show just how smart they are about this stuff Like they did the really great questions. Oh, yeah. Yeah. Yeah. Yeah. Yeah. It's like. Oh my gosh We should be handing out degrees Except for this first guy this first guy's an idiot. No, I'm just kidding

No, okay, so what you got for me? This is Farouk from Uzbekistan writing from Norwich, UK It's my favorite movie interstellar after I've seen your interview with Kip Thorne There's something that bothers me now When they were on Miller's planet with giant water waves next to the black hole being immensely time-dilated Relative to one of their crew that remain behind on the ship and aged faster What if they had?

Telescopes and looked through the windows of the ship at their crewmate aging years for their minutes wouldn't they be able to see the faster passage of time or Would it have well or what would it have looked like or is light itself? Affected and won't allow this to happen So if you had a telescope and look down at them near the surface of the black hole everything would be going in slow motion I understand but it's also kind of insensitive with the look into the telescope. Hey guys. Look at this

Look at he's aging so fast. He just he just took out a reverse mortgage on his house

While we watch it yeah, I I just had a sandwich and during that time he's been divorced three times And he doesn't have any visitation rights like what? They would look down and see them moving slowly and when you're near a black hole you see the rest of the universe Unfold in rapid speed, and so yeah, you it's there's nothing special about the light that happens It arrives to you at a different pace Depending on how deep into a gravitational well you are and the reason why that orbit was chosen to be very close to the black hole

It was to maximize the time dilation It's called time dilation when you create this difference in the rate of time by the way that exists for our GPS satellites We are closer to the center of the earth than the GPS satellites are so we Experience slower time than they do yet. We get our precise time

from them So when they send down their time from orbit they Pre-corrected for Einstein's general theory of relativity so that our time works at our location on Earth's surface

Yeah, so it but you in use and saying you looking through the telescope. You're seeing things in slow motion Essentially, is that if you're looking at at their behavior on the planet? Yeah. Oh, yeah because it because even if the guy Because light itself gets stretched and so it looks like he's buffering basically is that what we're saying

That's an interesting word to bring into relativity. I have to think about that So no just the timeline. So here's the thing. You are a prisoner of the present For forever transitioning between your inaccessible past and your unknowable future That's all you know about your own timeline. There are other timelines That are proceeding at different paces than your timeline depending on how fast they're moving in space Relative to you and depending on how much gravity they're subjected to so time is not absolute

That's that this was the big discovery of Einstein Isaac Newton is turning in his grave upon hearing this If you could say who's president we it's basically like a who's present are we talking about right does this? Relativity does this mean there's no shared Now so the idea of watching someone age across frames is already kind of the wrong question in a way You're right. Some of those questions just kind of lose meaning like what is the now for everybody if your timeline is

Unfolding more slowly than mine Right because you can't watch them age and fast forward because the universe won't let you skip ahead to the consequence It won't let you skip ahead, but you can watch a timeline unfold faster than yours and if time is that dilated does that mean? You couldn't actually watch him aging and fast forward because the light carrying that information is Stretched and delayed as well. So the universe you put yeah, I think if you're tracking time These stretching of the so rather than think of stretching of the light think of light pulses

Right and so what's the interval between pulses? And that's a nice way to sort of tick out the time Mm-hmm. The pulses you would see down on the on the planet. I guess what's called Miller's planet I just remember the

name of it. Yes, no wish if you all agreed Let's pulse every second and then you see their pulses down there and their pulses happen every two or three seconds Even though according to their watches, it's every one second so it's not that the light stretches so much as the pace at which time unfolds shifts and

So yeah, and and they did it right. We did it right Mark Phillips florescent Missouri. Hi, Dr. Tyson. This is Mark

Florescent F-O-F-L-O-R-I-S-S-A-N or florescent Okay, if we are living in a in a simulation Would you want to know and Would you choose to embrace your place or attempt to break the cycle and communicate to the overlords knowing you Neil You'll need to communicate because it's gone a lot of that

I love that question and what would and third part of the question What would you wish to gain from contact with them or have you already? And you are just controlling us. I'm not authorized to comment

That's a great question. I love it. I embrace all objective truths No matter what I mean, it's not that long ago in the era of our grandparents perhaps Where it was perfectly acceptable for the doctor to lie to you To tell you you don't have terminal cancer and no, you're not gonna be dead in six months They that was somehow accepted

In fact, I think there was such a scene in was it a streetcar named desire. What's the story that has big daddy in it? Oh, that's mendacity. That's a cat on a hot tin roof. Well a cat on hot tin the cat on hot I don't know why confused those two so cat on a hot tin roof big daddy or whatever his name was he He was told everyone else knew he was gonna die and the doctor told him he was like just fine To me that is today completely unacceptable at least for me. I want to know all truth so that I can

reckon With that truth in whatever way I see fit who live a lie is To not fully be in touch with all that

Controls your life all that matters all that so yeah, maybe that's because I'm a scientist. Some people just say lie to me Lie to me. There's a lot of

politics, right? Lie to me. Tell me there is no global warming. Tell me we're not right Tell me just lie to me and I know everything will be better and and that's okay Temporarily and then it catches up with you and by then it's too late Yes, so yeah, I would want to know if I was in a simulation. I I agree. I would want to know B

I totally want to talk to the overlords Absolutely, I Want to talk to the manager. I have questions

First of all, who picked my avatar? Why don't I have a better body? That would be one You had infinite rendering power and you chose to make me a guy who pulls a hamstring tying his shoe That would be number two Why did you give me mild acid reflux? I have a lot of issues with this overlord person and with our luck It's a 14 year old kid who's gets like a triple-a plus an alien kid in his parents basement. Yeah Listen if we're in a simulation, I just want the patch notes. That's all I want

Okay, the programmer of the simulation in which we live gave you freedom to join a fitness center

It is obvious to me you've never stepped into a fitness center Okay, this is why I Don't want to know all truths. This is why there are some there are some things I don't want to know So what would you change? No, no questions. What would I ask and I would I would ask is every bit of this universe?

Noble through laws of physics Or is there something you just got lazy programming and it will never make sense to anybody and no one can never create a Law for it because you just pulled it out of your ass That's what I'd want to know because if they didn't then I would still continue to explore With the laws well, it's funny because in my mind when I read this question before

We came on the air it was like it's sort of maybe the the other side of the coin of what you just said which is Is there only a simulation or is there some Being or power the femoral ethereal that has some hand in how we exist Does that I don't know that we would ever be able to deduce the difference well, but could you that's the point, right? So the simulation

Maybe only exists because of this greater power being right? We don't you'll never know the answer

Or is the greater powerful being? Yeah, but that's deal with that. That's yeah, but but we don't know what the sim is a simulator or a human being Is it a is it an alien of some kind like what is who is the simulator? It doesn't matter. I mean I'm saying it's someone vastly Smarter than us. That's a whoa. It's like an answer. Speak for yourself, buddy. Mr. insecure low IQ guy I Concap first of all in this one question you told me I have arms like wimped spaghetti noodles

In terms of my body's height and I'm dumb as a stump. Is this how we're gonna do the whole show Arms I didn't talk about your brain. You have a brilliant brain

No, no, so that's what I would do I would say shall I continue as a scientist Exploring the no ability of the universe or is there some point where you just gave up and it's all random That's what I want to know. I believe here. It has good precedent There was a day before Newton you'd look up and the planets would move against the background stars and the moon and the sun and no One really understood it we could sort of trace it But an understanding eluded us

But that was okay because the sky was the realm of God and how could you possibly? understand the mind of God and then Isaac Newton says here it is there's three equations and

Some accused Newton of heresy for for Stepping where previously only God had tread and so in this case is is God a deity as known through religious texts or is God a programmer in an alien basement in either case

Whatever is going on is created by them and is vastly more complex than we are and we're just Inside of it just poking well What is it? I mean if it's so sophisticated that it can create all this one question I would have in all seriousness is why have you created so many imperfections when you have this amazing ability to create this Simulation, okay, so I have an answer to that. I have an answer. Okay. Okay, just when everything is going fine Just when everything is going fine

The programmer gets bored. Let's say let's throw in COVID

Okay, I know let's throw in the feature where Paul has to get up three times a night to pee at How about that? Let's throw that in there just to make it like you know, it's TMI Paul I don't need to know about your bladder conditions So then we finally get through COVID. Okay, and we have the vaccine better then a

Billionaire New York real estate developer becomes the most powerful person in the world

If there were ever evidence that an alien programmer were in charge just for their entertainment That Hi, I'm Ernie Carducci from Columbus, Ohio I'm here with my son Ernie because we listen to star talk every night and support star talk on Patreon

So Paul here's an interesting fact Hmm if we reach the point Where in our exploration of the universe where it's all random because the programmer got lazy, right? That's that's interesting At that point science loses all utility Science exists only because something I figure out here applies over there

And I can predict its behavior and understand it and manipulate it and use it in the service of my own survival and and if If it were all random there would be no such thing as science Well, but no, but you couldn't you but couldn't you use science to help explore and try to explain the randomness and the chaos?

Seriously, so so you want you want a scientist to say okay, paul today Your whole day will be chaotic. I predict that go forth What what are you expecting? No, no, no, no, but you can't can't you make the argument that you could use science to sort of explain Chaos, I mean isn't isn't that there truly random? I mean the chaos is a very specific thing and in mathematical mathematically and the interesting thing about chaos is

You can predict with precision how uncertain certain quantities will become over time And you can watch the chaotic regime unfold This is what makes weather predicting so challenging because there's so many variables the heat from the sun the absorption of the earth the reflectivity of the earth the

The oceans which are a different temperature from the land how much moisture is in the air All of these factors come together

And the more they come together and the more in the future you try to understand it the more chaotic it becomes So we try to get handles on it. That's why they say Uh, the models we have this model and we have that model and here's where they agree So you go where they agree To try to boost your chances of understanding The objective reality of the system

But i'm just saying if if if if no laws Discovered here apply over there And it doesn't matter where here is and it doesn't matter where there is Then that undermines the entire profession of science Right and may sit on the fact that I tell you that on earth as it is in the heavens Laws of physics on earth applied to the moons planets stars and galaxies

Not only across space, but through time. That is not an assumption That is a measurement we have made And that's extraordinary Is any reason is there a tablet in the sky that says that must be so I don't know I'm delighted. We live in a universe that it is so Absolutely. All right, let's move on to the next question. This is uh, mackayle from canada since it's around three kelvin in space

Could an asteroid be a superconductor? And if so Would it get stuck if it? If it went too close to a huge magnetic field like the suns or ours

There you go I love that question very clever. This what's the person's name? This is mackayle mackayle So thank you for that question. Mackayle knows that metals certain metals and other materials When you reduce the temperature in which they're immersed the behavior electron changes so that they no longer act separately

from one another As the temperature drops as they get cooler The wavelength of each electron grows bigger and bigger and bigger Until it reaches i'm simplifying this but it's it's basically what's happening Until the temperature is so low By the way, this is a quantum. It's a quantum phenomenon. Was that absolute zero? Is that absolute zero?

There's three degrees above absolute zero, right? So no more. He can be taken out When it gets to zero, right? So as you drop the temperature the wavelength of the electrons gets bigger and bigger Quantum mechanics dictates this to a point where the wavelength is so large all the electrons Behave as one electron Because their waves overlap and synchronize and when that happens electrical current passes through

without any resistance at all And so it's called superconductivity At three degrees ought to get some superconductivity in some metals. I haven't checked the chart lately That who goes superconducting at what temperature? Some go superconducting at even cooler than three degrees kelvin So this let's start with that. So great question

I'd have to check my numbers to see who would be superconducting at those temperatures a b It's not entirely three degrees If you're near a star So the asteroids are orbiting the sun many of them are very dark Sunlight gets absorbed raising their temperature So you're only going to hit three degrees like an intergalactic space

We are so far away from nearby stars that this very little heat Added to your to your to your rock temperature jason ashton from liverpool liverpool The beetles are from liverpool. Yeah, no, yes Wait, wait, if he's from liverpool, you have to read it with a liverpool accent. Oh, please

Jade That's like irish or something. Yeah, oh The home of the beetles. Yeah I like you. You know, you know, I had paul mccarney on my podcast, you know Along with you. I do know that. Okay. Let's move on I've been a patreon member for over a year and finally have a question that is probably more basic than uh,

Then my and your expectations are so i'm feeling a little bit dumb here if the moon left earth's orbit Would the oceans no longer? Have tides They would still have tides caused by the sun Yes The sun's tides are about one third the strength

Of the moon's tides or to put more precisely the whatever is the height of the tide that you measure A third of that height is attributable to the sun. Yeah, it's smaller less dramatic It's like what I like to call decaf get gravity.

It's like That works that works and what's more important here and by that's a perfectly legitimate question. I love it and so In fact, there's a whole book

Written by a colleague of mine called it. What if there were no moon and then the whole book just goes talks about the consequences of that And we probably wouldn't have moons

After the moon

Yeah, yeah, we have some other sort of reckoning of time So the moon has the dominant tidal bulge. All right, so there it is and the earth turns in and out of that bulge That bulge is always there However And it's always in the same place Well, it's a little bit ahead of the moon in the moon's orbit

All right, but we rotate faster than moon's orbit around us So we rotate into and out of the tidal bulge And because the moon still moves around us high tide shifts each day by about 45 or 50 minutes Because of this effect It's all one Synchrony of time space the effect of the tidal bulge being slightly ahead of us

It's slightly ahead of the moon, but the moon is orbiting the earth. So how far does the moon go in a day? It goes about You know one, you know 45 50 minutes worth Of your clock time Around the earth So that therefore the the tidal bulge has shifted by that much here, but that's not even my point

The pole just the bulge is always there When we line up with the sun The sun's high tide adds to the moon's high tide And you get full moon high tides So when people think the full moon has some extra tidal forces on us

No, it's the sun is the sun lining up added to the moon's tide and the moon's tide is just always there But the sun adds to it. And so it's a delusion. We think oh the rays of the sun the moon We have werewolves. We have all life Anthropic stories that have been written. We've got all of this About moonlight And plus the moon is only reflected sunlight. So if the werewolf turns into werewolf

Under full moon it should definitely do it under sunlight And a lot faster and with a lot more air Bigger teeth bigger fangs But what so what is the remind us the full moon High tide which means sort of the sun and the moon have sort of aligned That happens every how often what is the frequency of that?

Once a month Well, actually twice a month. So so it'll work for full moon Because earth moon and sun are lined up and it works for new moon where

the moon is on the other side And you get the tidal bulge So you get the highest of the high tides during new moon and full moon So people aren't thinking that the new moon has special powers even though it's making tides just as high as full moon

Because you can't see the new moon So people are very influenced by what they see and what they think is true Rather than all that they can't see and what we otherwise know is true

Yeah, that is Yeah, there's a lot there

Hello from flesherton, ontario Not going to be a part of the 51st state canada All right But we need your we need your health care We need your health care your your pleasant qualities your lovely manner with people your

Your maple syrup we've only got a couple of states that really generate that come on from on new york That's it. That's from new hampshire. Yeah, it's gonna be some main too. Um, yeah Neil my 13 year old son was so blown away by our experience at your show He wanted me to pass on this question. Why are your t-shirts so expensive after the show? No, that's not the question I don't have some t-shirts after this. I have no merch. I mean people said I should but I there's no merch You sell those Neil deGrasse Tyson fit you sell those Neil deGrasse Tyson fake mustaches for 15 each

I know you do All right, Neil my 13 year old son was so blown away by our with our experience at your show He wanted me to pass on this question. Could you ask if he thinks your memories and consciousness are made from adams? Love you guys and keep the laughs coming andrew nisker So this is my favorite joke that's safe for elementary school Okay

Never trust adams They make up everything

That's safe for like third graders We're good there. So Yeah, I mean We we so want to believe That there are forces and things going on that defy account Or description or investigation by the laws of physics

That erds is very strong because oh, you'll never you might explain Uh, you know language or whatever, but you'll never explain love You know they and I say which is it because you don't want us to ever explain love Would you have some reason for thinking that this is really true? so A couple of things about our consciousness

Or our thoughts. Let's just talk with our our thoughts. We have come to learn That the brain operates with by electrochemical signals between the synapses of our nerve cells of the of the brain cells in your head, okay and

There it is By the way, the medical profession could not have known that Before the discovery and the harnessing of electricity Think about that So much of physics has fed the medical professions

powers to diagnose and treat the human body That if anyone says, okay, we're done with physics. Now. Let's invest all in medicine That yet that's there's a limit to how far that'll go given how much they've tapped

discoveries by my people in my fields, okay so when you know That that thoughts are communicated through electrochemical signals and then you Stick a probe in there and stimulate a part of the brain that you know behaves in a certain way And then you see the the person

react You kind of know what's going on there. This is as Adams said Yeah, Adams are the building blocks of every building blocks of cells and the cells of the building blocks of our neuro physiology Okay, so so but the real evidence here because you can't just experiment on people's brains that you know, really do that

Hang on a second. All right, let him go. We can't Let's you kids your kids in the box Yeah, get them out of the box and give them some peanuts and hit them to that We can't do the experiment. Go ahead. Sorry people who have many strokes Right. So a stroke of broken blood blood vessel. There's oxygen doesn't get to a part of the brain if there's many strokes

The identity of the person Begins to fade depending on what part of their brain Was touched by the stroke And so is it their facial recognition? Is it

their spatial recognition? Is it their appetite for food? Is it their ability to speak? Is it it is that they're hearing sight locomotion

If you have a series of mini strokes bits and pieces of you basically disappear So everyone who says what happens to your consciousness when you die A series of mini strokes are mini deaths of your brain and as far as we Can tell that's what's happening to you when you die. None of that works so Now that's not very comforting to be especially not too religious people have a whole story of what happens to you

When you die, all i'm saying is if you're going to ask a scientist to answer that question the point is Once your brain ceases to undergo electrochemical changes Then you don't exist anymore. Everything we know about you no longer exists So, I mean dementia is sort of in that family, right? So for example, correct. And that's not even a stroke

I I should have included but it's in that family of you know So my mother has it right now and you can see there's just pieces of it But the fascinating thing is her long-term memory is insanely accurate It's stored differently and possibly even in a different place I would introduce you to her and ten minutes say to she be like, how do you how do you know my son again?

But then she can in the same breath relay A story when she was 12 of spending a day with her father with great exactitude and so It is it the shifting of the atoms in the cells? Is it is it the shifting of the atoms of the cells? It's happening there. Is it the death of some atoms in cells? Like what what is the atoms don't have life or death? They just comprise larger molecules and the cells are composed of the molecules. So

You can't blame atoms directly. You have to you bring you you have to credit or blame the functioning of living Tissue Okay, that's really what it is. The atom is just an atom

now if I may I mean no disrespect, but I heard this joke told by another comedian Who said that that if you're a comedian being around people with dementia is Excellent because you could tell that's the same joke every ten minutes

absolutely People short-term memory dementia, right? It also helps with guilt because like you'll you can say to your mother Uh, well, I haven't seen you in a while. No, I was just here yesterday. Oh, that's right. You know what I mean? So like you don't have to feel so bad You know Yeah, exactly. It's like, oh, you're such a good son. I know

I was here yesterday and I painted the whole house point is atoms everything that is not empty space or energy in the form of photons represents As particles in the universe and certain combinations of particles we call atoms If you have a nucleus and at least one electron It's an atom

and so atoms comprise everything That's that's hence the joke don't never trust atoms. They make up everything If you go down to like a snowflake and a snowflake is six-sided snowflake It's a crystalline structure of the water molecule water is H_2O

You break that apart. You have two hydrogen atoms one oxygen atom. So atoms are everywhere But atoms don't have agency. They're just well, sorry They might want to combine with other atoms to make molecules, but That's it's a very simple. Are we mating or not? It's not otherwise Let me give Paul McCurrie dreams that happens based on life experience and left over

neurosynaptic history of your brain and Stuff here's unresolved trauma that might have happened in your life joys So yeah, atoms are everywhere

So Paul we got enough time for like a lightning round This is Todd Chambers from Yuba City, California in the system of the universe What is or are the purposes of a black hole? Do they recycle matter and energy to speak of purpose implies intent? And in physics, we generally don't think of the universe as having intent They're just laws and forces and things

flow according to those forces Guided by the laws of physics as we've come to describe them So you can ask what role they play And if you wander too close to a black hole, that's all she wrote You you'll get pulled in spaghetti

And You won't be able to tell anybody about it So to think of a role No, there's there's roles that nobody there's are no roles. There's just what it does And by the way, if the sun became a black hole today Earth would plunge into darkness

But we can still continue to orbit around it in just the same way we are right now We wouldn't be pulled in yeah black holes are not giant sucking machines But there's stronger gravitational force only at its surface Don't disagree with me. Don't you don't know what you're talking about. I'm not disagreeing with you. I'm just simply saying you're wrong

There's a difference Politicians will disagree Yes, okay But if one person is wrong, you're just wrong. It's not that we disagree

Okay, all right, we're gonna yeah, so it doesn't become a giant sucking machine It'll it's we speak of the high gravity of a black hole because of the high surface gravity Because the black hole is so small you get really close to its center and that's where the extreme gravitational phenomenon occurs Got it. Okay, uh, right here. We'll keep going here. Liam Corcoran. Hi neo This is Liam from Rhode Island my home state The best little state in the union. Is there anywhere in the universe?

That you could be outside the gravitational influence of any matter beside your own And would this be the closest you could get to experiencing pure spacetime? Totally unwarped by matter. I love that question I love that question There is Oh, I forgot what the metric is called. There's

There are approximations we make when we're doing some calculations We assume space has no matter or energy in it. So that's perfectly flat Okay, and then you can do some calculations more easily than if you admit that there are stars or planets or black holes Distorting that spacetime the mathematics becomes much more complicated if the spacetime is curved Then if it's flat, so now in 1968 if memory serves Walter Cronkite The anchor for CBS news was describing in december the voyage of Apollo 8 to the moon and he says

As of 9 22 this afternoon Apollo 8 astronauts have left the gravity of earth And I remember hearing I was like, what they're on their way to the moon

and last I check the moon orbits earth so So I know what he meant But that's not what he should have said Okay, what he meant was we reached a point where the moon's gravity

Pulling it in Is stronger than our gravity trying to come back to earth so that the moon's gravity takes over and it just falls towards the moon That's not the end earth gravity extends to infinity as does every other gravitational field You cannot escape The extent of gravity no matter what But it weakens over time the farther you get away from earth

It dissipates in strength the weaker and weaker it is but it doesn't it's never zero Right, that's true for everything the sun the galaxy Galaxy clusters everything my name is emily from berkeley. I am a first-time poster I hope this is an interesting question many times when people talk about Particle pairs being separated at outer surface of black holes Where one half of the pair is on the outside and one half is doomed to the inside

They will say that it should be theoretically possible to reconstruct the information That fell into the black hole. Why is it important to theoretically retain this possibility? Oh, this comes from information theory because there's a certain amount of information And if you fall into a black hole was the information destroyed That's the question and so there's a big debate

About information information theory is relatively new you know, it's sort of Well, it comes out of entropy so in the last hundred years or so and especially in the last

50 50 years puts it in the 1970s, I guess. Wow. Yeah, and if you Believe or want it to be so that information is never destroyed Then when the black hole later evaporates Which it will do by Hawking radiation. Well, that is excuse me. That is the way it evaporates. Okay, so so that particle pair

is conjured out of the gravitational field and the if you took an inventory Of the particles that appear out of the gravitational field of the black hole They will exactly match everything the black hole has ever eaten In its life. How how so? Well, what's freaky is it's not the same particle coming out that survived the trip in and survived the trip out

Particle that being created from the energy field of the black hole Energy is creating it and it has a memory of what was there before that's tantamount to preserving the information that went in so Now there's a little bit that I'm unsure of we're gonna have to bring Jana Levin on this If if you throw in a DNA molecule, there's a lot of information. There's your whole identity It's not going to come back out as a DNA molecule

in particle pairs No So where's the information it'll come back altered in some way now It won't come back as a molecule at all and come back as particles So how do I recover the information that the DNA molecule contained? As a representation of you. I don't have an answer for that. We've got to check with either one of the two bryans or

With jan 11 on that. We'll bring them back on. Don't worry about it. I got it. I got it. You got this Yeah, what happens is it goes in and there's centrifugal force that happens along with a certain level of Radiation red wave and then you get shit thing. All right moving on

Mercedes Dominguez. Hello Mercedes here from Denver, Colorado. I've heard of the cosmic web before My question is is that a real thing? And if so, what is it? Yeah, so it's just a word that we apply to How we find the galaxies configured in the universe so

Galaxies are not evenly separated. There's gravity operating and there were some fluctuations in the early universe That preloaded some parts of the universe relative to others when you step back and take a look at it all you see this sort of network of filaments and clusters and It says though you took a sponge

Cut it in half Look like a nice thick sponge cut in half and look at the textures and structures. There are places where the fibers meet There are places where it's empty It's almost like a honeycomb in a way not as symmetrical but not as organized Not structured correct. So the galaxy the galaxies aren't sprinkled evenly They they line up along huge strands of matter mostly dark matter with these like empty voids in between and it looks like a

three-dimensional spider web in a way Or or a sponge. Yeah, I mean cosmic web sounds a little more romanticized than cosmic sponge

Yeah, that's a little that's a tough sell to the uh, yeah But the point is the universe isn't random. It's organized. I mean, it's not like pinterest organized I mean gravity collects it. You want to call that organization with stuff piled into the corners of every Very possible. That's a good point. In fact Here's an interesting phenomenon if you look at a movie where they show a scene in the fall Basically, they're set designers scattering leaves onto the street. Okay, if it's not actually in the fall

What they always get wrong is They try to scatter the leaves evenly on the street It's never that the breeze comes by and it piles into one corner So mother's new leaves fall down But it's never evenly distributed. You can always tell because people don't understand random They think random is scattered everywhere random has clumps. That's how that is

So So I can always tell if they fake the street for that reason But the but science can't answer the the randomness question here, right? Like that's that's isn't that to me is the fascinating thing here is sort of it's sort of organized It's in a way organized chaos. Maybe that's not the right technical term Forces of gravity are operating on it

So there's not nothing more mysterious than that and dark dark matter is mysterious We don't know what that is, but that's operating on the gravitational fields Around which all of this is collecting. So we speak of that entire that entire map Of the universe as the cosmic web

So Paul that's all the time we have plus. I think I hear your mother calling you for lunch

It's great to be on with you. Yes, man, and keep us posted on the end of colbert or the end of days If the oh, you'll know because I'm I'm gonna be I'm coming out to your neck of the woods I'm doing I'm doing landscaping after this. I need to I need the money, buddy Can you buy a place and get a lawn so I can put it? I need the

Be a friend. Be a friend. Will you? All right, we'll work on it. All right All right, paul mercurial always good to have you ma'am. Yeah, always great chara chara chubby terry cruise This has been another installment of cosmic queries grab bag edition With paul mercurio kneel the grass tyson as always bidding you to keep looking up